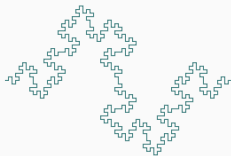


CoCoMA (3) – Game-theoretical Coordination

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CoCoMA | Master ANDROIDE

Outline of the Course

- Part I: A crash course on potential games
- Part II: Utility design
- Part III: Learning/Negotiation mechanisms

References for today's lecture

On potential games:

Monderer, Shapley. *Potential Games*. GEB-1996.

On wonderful life utility and utility design:

Wolpert, D., and Tumer, K.. *A Survey of Collectives*. 2004.

On utility design, learning and negotiation mechanism:

Arslan, Marden, Shamma. *Autonomous Vehicle-Target Assignment: A Game-Theoretical Formulation*. JDSMC-2007.

Marden, Wierman. *Distributed Welfare Games*. OR-2010.

A game-theoretical perspective on coordination

Setting: agents must find an assignment to tasks.

What is different from what we have seen before:

- several agents can share the same task
- utility for the social objective defined at the level of tasks
- but we only control the utility of agents!

The setting

- a set of agents $\{1, 2, \dots, n\}$
- a set of tasks $\{t_1, t_2, \dots, t_n\}$ + sometimes t_0 for the void task
- each agent can play an action from the set A_i (here the set of tasks, or a subset of tasks if perception is limited)

An action profile can thus be seen as a tuple a tasks selected by each agents, eg.

$$\langle t_1, t_1, t_3, t_4 \rangle$$

- each task generates some utility depending on the agents assigned to it. Denote by $ag(t, a)$ the agents who selected t in a , and $n_t(a) = |ag(t, a)|$. For instance, $ag(t_1) = \{1, 2\}$.
- the global objective (social welfare) is defined as

$$sw(a) = \sum_{t \in T} u_t(a)$$

Example

Two areas to fix in a space ship: t_1, t_2

One is crucial and rewards 10, as long as one robot is assigned to it. Another one is not critical for the safety of the spaceship and provides 2 as long as one robot is assigned to it.

Two robots available to perform these tasks.

Optimal social welfare is obtained when $\langle t_1, t_2 \rangle$ or $\langle t_2, t_1 \rangle$

	t_0	t_1	t_2
t_0	0	10	2
t_1	10	10	12
t_2	2	12	2

Denote by a_{-i} the action profile consisting of all other actions except that of agent i

Recall that a PNE is an action profile such that any agent plays a best response to the current action of other players, ie

$$u_i(a_i, a_{-i}) = \max_{a' \in A_i} u_i(a', a_{-i})$$

Better response update (BRU) for agent i : deviation leading to better utility for agent i

Potential games: A crash course

Potential games

A **potential game** admits a potential function Φ such that, for any $a_i, a'_i \in A_i$, for any $a_{-i} \in A_{-i}$:

$$u_i(a_i, a_{-i}) - u_i(a'_i, a_{-i}) = \Phi(a_i, a_{-i}) - \Phi(a'_i, a_{-i})$$

2, 2	0, 3
3, 0	1, 1

?	?
?	?

Ordinal potential functions

An **ordinal potential function** admits a potential function Φ such that

$$u_i(a_i, a_{-i}) - u_i(a'_i, a_{-i}) > 0 \Leftrightarrow \Phi(a_i, a_{-i}) - \Phi(a'_i, a_{-i}) > 0$$

A **generalized ordinal potential function** admits a potential function Φ such that

$$u_i(a_i, a_{-i}) - u_i(a'_i, a_{-i}) > 0 \Rightarrow \Phi(a_i, a_{-i}) - \Phi(a'_i, a_{-i}) > 0$$

Generalized ordinal potential games exactly correspond to games with the finite improvement property, ie. any sequence of BRU terminates

Example

Can we find a potential function?

2, 2	0, 3
3, 0	1, 2

?	?
?	?

Example

Can we find a potential function?

2, 2	0, 3
3, 0	1, 2

?	?
?	?

No. But we can find set an ordinal potential function.

Example

Can we find an ordinal potential function?

1, 0	2, 0
2, 0	0, 1

?	?
?	?

Example

Can we find an ordinal potential function?

1, 0	2, 0
2, 0	0, 1

?	?
?	?

No. But we can find set an generalized ordinal potential function.

Utility Design

The challenge

How can I design the utility function of the agents so that a global optimum is reached?

So that:

- convergence to a PNE is guaranteed
- budget is **balanced**: for any t , the utility distributed to agents assigned to t is exactly equal to what the task generates, ie

$$\sum_{i \in ag(t,a)} u_i(a) = u_t(a)$$

- communication requirements of agents remains low.

Communication requirements

Marden et al. distinguishes 3 levels of information requirements:

- **High** – the distribution to agent i given player set S requires knowledge of welfare of all player sets $T \subseteq S$
- **Medium** – the distribution to agent i given player set S requires knowledge of welfare S and $S \setminus \{i\}$
- **Low** – the distribution to agent i given player set S requires knowledge of welfare of the player set S

Fully aligned utilities

Simply let the utility of agents be exactly the global objective

$$u_i = U \text{ for all agents } i$$

Some remarks:

- in that case U can play the role of a potential function
- this guarantees the social optimum to be a PNE
- but there may be other PNE
- **learnability issue**: an agent's own action may not be significant enough
- not budget balanced

Example

		Agent 2		
		t_0	t_1	t_2
Agent 1	t_0	(0,0)	(10,10)	(2,2)
	t_1	(10,10)	(10,10)	(12,12)
	t_2	(2,2)	(12,12)	(2,2)

Equally shared utilities (ES)

Equally divide the utility of the task among agents assigned to it. Denoting $n_t(a)$ the number of agents selecting t in the action profile a

$$u_i(a) = \frac{u_t(a)}{n_t(a)}$$

Some remarks:

- convergence is not guaranteed in general
- special case where convergence is guaranteed: anonymous games (agents are not distinguished by tasks)

Example

		Agent 2		
		t_0	t_1	t_2
Agent 1	t_0	(0,0)	(0,10)	(0,2)
	t_1	(10,0)	(5,5)	(10,2)
	t_2	(2,0)	(2,10)	(1,1)

Observe that sw optimal allocations are not in equilibrium.
Dynamics will converge to (5, 5), the only equilibrium.

Marginal Contribution aka Wonderful life utility (WLU)

Idea: align utilities with marginal contribution of agents

$$u_i(a_i, a_{-i}) = U(a_i, a_{-i}) - U(a_0, a_{-i})$$

where a_0 is the null or default action of agent i

Some remarks:

- in that case U can play the role of a potential function
- this guarantees the social optimum to be a PNE
- but there may be other PNE
- what exactly means the null/default action can be a matter of discussion: not doing anything, ...

Example

		Agent 2		
		t_0	t_1	t_2
Agent 1	t_0	(0,0)	(0,10)	(0,2)
	t_1	(10,0)	(0,0)	(10,2)
	t_2	(2,0)	(2,10)	(0,0)

Here the default action is selecting no task, ie. t_0

Some Limitations of Wonderful life utility

- not budget-balanced!
- the approach may not be appropriate in **tightly coupled domains**, ie. when agents need to coordinate with several other agents, and have little opportunity to get utility otherwise
- the requirement that agents can compute $U(a_i, a_{-i})$ is too strong and we have to rely on an **approximate** of WLU instead

Some examples:

Douchan, Kaminka. *Swarms can be rational*. AAMAS-19.

Rahmattalab, Chung, Tumer. *D₊₊: Structural Credit Assignment in Tightly Coupled Multiagent Domains*. AAMAS-20.

Shapley values contributions (SV)

Recall (one of the possible) definition, in terms of possible orderings Π , where $mc(\pi, i)$ stands for the marginal contribution that i brings to the current coalition in the order π by playing a :

$$u_i(a) = \frac{1}{n!} \sum_{\pi \in \Pi} mc(\pi, i, a)$$

- guarantees budget-balance
- note: similar to ES when agents are anonymous

Other examples

$u_t(S) = 1$ as soon as $|S| \geq 1$, and 0 when $S = \emptyset$

- Wonderful life utility:

Other examples

$u_t(S) = 1$ as soon as $|S| \geq 1$, and 0 when $S = \emptyset$

- Wonderful life utility: 0 for each agent, unless alone to do t
- Shapley value (=ES):

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$u_t(S) = 1$ when $S = N$, and 0 otherwise

- Wonderful life utility:

Other examples

$u_t(S) = 1$ as soon as $|S| \geq 1$, and 0 when $S = \emptyset$

- Wonderful life utility: 0 for each agent, unless alone to do t
- Shapley value (=ES): $\frac{1}{n}$ for each agent

$u_t(S) = 1$ when $S = N$, and 0 otherwise

- Wonderful life utility: 0 for each agent, unless everyone does t
- Shapley value (=ES):

Other examples

$u_t(S) = 1$ as soon as $|S| \geq 1$, and 0 when $S = \emptyset$

- Wonderful life utility: 0 for each agent, unless alone to do t
- Shapley value (=ES): $\frac{1}{n}$ for each agent

$u_t(S) = 1$ when $S = N$, and 0 otherwise

- Wonderful life utility: 0 for each agent, unless everyone does t
- Shapley value (=ES): $\frac{1}{n}$ for each agent

It is common to call:

- **Price of Stability**: the ratio between the best PNE and the social optimum
- **Price of Anarchy**: the ratio between the worst PNE and the social optimum

What we have seen so far is that (often) $\text{PoS}=1$ in our setting but that PoA can be > 1

(See AOTJ!)

Dynamics

Learning as a negotiation mechanisms

As there are multiple equilibria, how do agents converge (if they do) to a specific one?

Can be seen as agents negotiating the assignment of tasks through an iterative mechanism

Several dynamics can be studied:

- each agents play simultaneously
- sequential: round-robin or random selection of next agent to play, etc.
- until convergence to an agreement (here: a PNE)

We will present the simultaneous versions but the others are easily derived

Myopic Best Response

- simply play best response to the actions of the other agents played in the last round

Action-based Fictitious Play

We can see fictitious play as an iterated negotiation where agents play simultaneously

- at each step of the negotiation, agents select a task
- let $n_j(k, i)$ be the number of times that i selected t_j up to step k
- this provides an empirical frequency for agent i
- play best response (in expectation) to the empirical frequency of the other agents

See [this](#) notebook for examples!

Regret Matching

- for each agent, maintain the cumulative regret of each task t (ie. the difference between the actual score obtained by i and the score he would have obtained by playing t throughout the different rounds)
- play according to mixed strategy proportional to positive regrets

See [this](#) notebook for examples!

Spatial Adaptive Play

- proposed by Young, similar to softmax and Gibbs sampling
- play a_i with probability

$$\frac{e^{\beta u_i(a_i, a_{-i})}}{\sum_{a'_i \in A_i} e^{\beta u_i(a'_i, a_{-i})}}$$

- β is the exploration parameter
 - if $\beta = 0$ agent i plays with uniform probability
 - as β augments the mass is put on best responses
 - β can be set so as to vary with time, eg. $\beta(t) = \log(1 + t)$